

1. Customer Review Cleaning System

```
import pandas as pd
import re
import nltk

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer

# Download NLTK resources
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')

# -----
# 1. Load IMDb 50K review dataset
# -----

df = pd.read_csv("IMDB Dataset.csv")

print("Dataset Shape:", df.shape)
print("\nFirst 5 rows:")
print(df.head())

# -----
# 2. Inspect missing values and duplicate reviews
# -----

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Reviews:", df['review'].duplicated().sum())

# Remove duplicate reviews
df = df.drop_duplicates(subset='review')

print("\nShape after removing duplicates:", df.shape)

# -----
# 3-7. Text preprocessing
# -----
```

```
stop_words = set(stopwords.words('english'))
stemmer = PorterStemmer()
```

```
def clean_review(text):
```

```
    # 3. Convert to lowercase
```

```
    text = text.lower()
```

```
    # 4. Remove HTML tags
```

```
    text = re.sub(r'<.*?>', '', text)
```

```
    # Remove URLs
```

```
    text = re.sub(r'https?://\S+|www\.\S+', '', text)
```

```
    # Remove numbers
```

```
    text = re.sub(r'\d+', '', text)
```

```
    # Remove punctuation and special characters
```

```
    text = re.sub(r'^a-zA-Z\s]', '', text)
```

```
    # Remove extra spaces
```

```
    text = re.sub(r'\s+', '', text).strip()
```

```
    # 6. Tokenize
```

```
    tokens = word_tokenize(text)
```

```
    # 5. Remove stopwords
```

```
    tokens = [word for word in tokens if word not in stop_words]
```

```
    # 7. Apply stemming
```

```
    tokens = [stemmer.stem(word) for word in tokens]
```

```
    # Convert tokens back into a sentence
```

```
    cleaned_text = ' '.join(tokens)
```

```
    return cleaned_text
```

```
# -----
```

```
# 8. Store cleaned reviews in a new column
```

```
# -----
```

```

df['cleaned_review'] = df['review'].apply(clean_review)

# -----
# 9. Compare original and cleaned reviews
# -----

print("\nOriginal vs Cleaned Reviews:\n")

for i in range(5):
    print("Original:")
    print(df.iloc[i]['review'])

    print("\nCleaned:")
    print(df.iloc[i]['cleaned_review'])

    print("\n" + "-" * 80 + "\n")

# -----
# 10. Export preprocessed dataset as CSV
# -----

df.to_csv("IMDB_preprocessed.csv", index=False)

print("Preprocessed dataset saved as IMDB_preprocessed.csv")

```

OUTPUT :

Dataset Shape: (50000, 2)

First 5 rows:

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

Missing Values:

```

review      0
sentiment   0
dtype: int64

```

Duplicate Reviews: 418

Shape after removing duplicates: (49582, 2)

Original vs Cleaned Reviews:

Original:

One of the other reviewers has mentioned that after watching just 1 Oz episode you'll be hooked. They are right, as this is exactly what happened with me.

The first thing that struck me about Oz was its brutality and unflinching scenes of violence, which set in right from the word GO. Trust me, this is not a show for the faint hearted or timid. This show pulls no punches with regards to drugs, sex or violence. Its is hardcore, in the classic use of the word.

It is called OZ as that is the nickname given to the Oswald Maximum Security State Penitentiary. It focuses mainly on Emerald City, an experimental section of the prison where all the cells have glass fronts and face inwards, so privacy is not high on the agenda. Em City is home to many..Aryans, Muslims, gangstas, Latinos, Christians, Italians, Irish and more....so scuffles, death stares, dodgy dealings and shady agreements are never far away.

I would say the main appeal of the show is due to the fact that it goes where other shows wouldn't dare. Forget pretty pictures painted for mainstream audiences, forget charm, forget romance...OZ doesn't mess around. The first episode I ever saw struck me as so nasty it was surreal, I couldn't say I was ready for it, but as I watched more, I developed a taste for Oz, and got accustomed to the high levels of graphic violence. Not just violence, but injustice (crooked guards who'll be sold out for a nickel, inmates who'll kill on order and get away with it, well mannered, middle class inmates being turned into prison bitches due to their lack of street skills or prison experience) Watching Oz, you may become comfortable with what is uncomfortable viewing....thats if you can get in touch with your darker side.

Cleaned:

one review mention watch oz episod hook right exactli happen first thing struck oz brutal unflinch scene violenc set right word go trust show faint heart timid show pull punch regard drug sex violenc hardcor classic use word call oz nicknam given oswald maximum secur state penitentari focus mainli emerald citi experiment section prison cell glass front face inward privaci high agenda em citi home mani aryan muslim gangsta latino christian italian irish scuffl death stare dodgi deal shadi agreement never far away would say main appeal show due fact goe show dare forget pretti pictur paint mainstream audienc forget charm forget romanc oz mess around first episod ever saw struck nasti surreal say readi watch develop tast oz got accustom high level graphic violenc violenc injustic crook guard sold nickel inmat kill order get away well manner middl class inmat turn prison bitch due lack street skill prison experi watch oz may becom comfort uncomfort view that get touch darker side

-

Original:

A wonderful little production.

The filming technique is very unassuming- very old-time-BBC fashion and gives a comforting, and sometimes discomforting, sense of realism to the entire piece.

The actors are extremely well chosen- Michael Sheen not only "has got all the polari" but he has all the voices down pat too! You can truly see the seamless editing guided by the references to Williams' diary entries, not only is it well worth the watching but it is a terrificly written and performed piece. A masterful production about one of the great master's of comedy and his life.

The realism really comes home with the little things: the fantasy of the guard which, rather than use the traditional 'dream' techniques remains solid then disappears. It plays on our knowledge and our senses, particularly with the scenes concerning Orton and Halliwell and the sets (particularly of their flat with Halliwell's murals decorating every surface) are terribly well done.

Cleaned:

wonder littl product film techniqu unassum old time bbc fashion give comfort
sometim discomfort sens realism entir piec actor extrem well chosen michael
sheen got polari voic pat truli see seamless edit guid refer william diari
entri well worth watch terrificli written perform piec master product one great
master comedi life realism realli come home littl thing fantasi guard rather
use tradit dream techniqu remain solid disappear play knowledg sens
particularli scene concern orton halliwell set particularli flat halliwell mural
decor everi surfac terribl well done

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Original:

I thought this was a wonderful way to spend time on a too hot summer weekend, sitting in the air conditioned theater and watching a light-hearted comedy. The plot is simplistic, but the dialogue is witty and the characters are likable (even the well bread suspected serial killer). While some may be disappointed when they realize this is not Match Point 2: Risk Addiction, I thought it was proof that Woody Allen is still fully in control of the style many of us have grown to love.

This was the most I'd laughed at one of Woody's comedies in years (dare I say a decade?). While I've never been impressed with Scarlet Johanson, in this she managed to tone down her "sexy" image and jumped right into a average, but spirited young woman.

This may not be the crown jewel of his career, but it was wittier than "Devil Wears Prada" and more interesting than "Superman" a great comedy to go see with friends.

Cleaned:

thought wonder way spend time hot summer weekend sit air condit theater watch
light heart comedi plot simplist dialogu witti charact likabl even well bread
suspect serial killer may disappoint realiz match point risk addict thought
proof woodi allen still fulli control style mani us grown love laugh one woodi
comedi year dare say decad never impress scarlet johanson manag tone sexi imag
jump right averag spirit young woman may crown jewel career wittier devil wear
prada interest superman great comedi go see friend

-

Original:

Basically there's a family where a little boy (Jake) thinks there's a zombie in his closet & his parents are fighting all the time.

This movie is slower than a soap opera... and suddenly, Jake decides to become Rambo and kill the zombie.

OK, first of all when you're going to make a film you must Decide if its a thriller or a drama! As a drama the movie is watchable. Parents are divorcing & arguing like in real life. And then we have Jake with his closet which totally ruins all the film! I expected to see a BOOGEYMAN similar movie, and instead i watched a drama with some meaningless thriller spots.

3 out of 10 just for the well playing parents & descent dialogs. As for the shots with Jake: just ignore them.

Cleaned:

basic famili littl boy jake think zombi closet parent fight time movi slower soap opera suddenli jake decid becom rambo kill zombi ok first go make film must decid thriller drama drama movi watchabl parent divorc argu like real life jake closet total ruin film expect see boogeyman similar movi instead watch drama meaningless thriller spot well play parent descent dialog shot jake ignor

-

Original:

Petter Mattei's "Love in the Time of Money" is a visually stunning film to watch. Mr. Mattei offers us a vivid portrait about human relations. This is a movie that seems to be telling us what money, power and success do to people in the different situations we encounter.

This being a variation on the Arthur Schnitzler's play about the same theme, the director transfers the action to the present time New York where all these different characters meet and connect. Each one is connected in one way, or another to the next person, but no one seems to know the previous point of contact. Stylishly, the film has a sophisticated luxurious look. We are taken to see how these people live and the world they live in their own habitat.

The only thing one gets out of all these souls in the picture is the different stages of loneliness each one inhabits. A big city is not exactly the best place in which human relations find sincere fulfillment, as one discerns is the case with most of the people we encounter.

The acting is good under Mr. Mattei's direction. Steve Buscemi, Rosario Dawson, Carol Kane, Michael Imperioli, Adrian Grenier, and the rest of the talented cast, make these characters come alive.

We wish Mr. Mattei good luck and await anxiously for his next work.

Cleaned:

petter mattei love time money visual stun film watch mr mattei offer us vivid portrait human relat movi seem tell us money power success peopl differ situat

encount variat arthur schnitzler play theme director transfer action present
time new york differ charact meet connect one connect one way anoth next person
one seem know previou point contact stylishli film sophist luxuri look taken
see peopl live world live habitat thing one get soul pictur differ stage loneli
one inhabit big citi exactli best place human relat find sincer fulfil one
discern case peopl encount act good mr mattei direct steve buscemi rosario
dawson carol kane michael imperioli adrian grenier rest talent cast make
charact come aliv wish mr mattei good luck await anxious next work

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Preprocessed dataset saved as IMDB_preprocessed.csv